

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PhD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Tutorial 7

Case Study: Demand Forecasting for a Consumer Goods Brand: Model Selection and Forecast Combination

Application area: *Marketing — Brand Management and Demand Forecasting*

Prof. Pedro Valls Pereira

`pedro.valls@fgv.br`

M5 Forecasting Competition data (<https://www.kaggle.com/competitions/m5-forecasting-accuracy/data>): daily unit sales of Walmart products 2011–2016 (requires free Kaggle account). Alternatively, the `Mcomp` R package (`install.packages("Mcomp")`) contains 3003 time series from the M3 competition, many of which are retail/marketing series.

Contents

1	Background and Objectives	2
2	Data Preparation	2
3	Exercises	2
3.1	Exploratory analysis	2
3.2	Model estimation	2
3.3	Forecast evaluation	3
3.4	Diebold–Mariano test	3
3.5	OLS forecast combination	3
3.6	Business interpretation	3
4	Core References	4

1. Background and Objectives

Accurate demand forecasting is the foundation of inventory management, promotional planning, and supply chain optimisation. This tutorial evaluates multiple forecasting methods using a real consumer goods series, applies the Diebold–Mariano test to compare accuracy, and implements forecast combination.

Learning objectives: (i) Apply ARIMA, ETS, GETS/Autometrics, and a simple ML method; (ii) evaluate forecasts using RMSFE, MAE, and MAPE; (iii) test equal predictive accuracy with the Diebold–Mariano test; (iv) construct equal-weight and OLS-combined forecasts; (v) interpret results for a marketing decision context.

2. Data Preparation

```
library(Mcomp); library(forecast); library(gets)
# Extract a quarterly retail series from M3
data(M3)
retail_series <- M3[[1400]] # quarterly retail sales example
y_train <- retail_series$x # in-sample
y_test <- retail_series$xx # out-of-sample (18 periods)
h <- length(y_test)
```

3. Exercises

3.1. Exploratory analysis

Plot the series. Apply STL decomposition (`stl` in R) to separate trend, seasonality, and remainder. Is the series seasonal? Does the trend appear deterministic or stochastic? Apply ADF and KPSS to the seasonally adjusted series.

3.2. Model estimation

Estimate the following models on the training sample:

- `auto.arima`: automatic ARIMA with BIC.
- `ets`: automatic exponential smoothing (ETS) model selection.
- Random walk with drift: $Y_{T+h|T} = Y_T + h\hat{\mu}$.
- Simple average of ARIMA and ETS forecasts (equal-weight combination).

For each model produce h -step-ahead point forecasts for the test period.

3.3. Forecast evaluation

Compute RMSFE, MAE, and MAPE for each model relative to the test observations. Arrange results in a table ranked by RMSFE. Does the combination outperform individual models?

3.4. Diebold–Mariano test

Apply the Diebold–Mariano test (`dm.test` in R's `forecast` package) to compare:

- (a) ARIMA vs. random walk (one-sided: ARIMA is more accurate).
- (b) ETS vs. ARIMA (two-sided test of equal accuracy).
- (c) Combination vs. best individual model.

Use HAC standard errors with bandwidth $h - 1$. Report p -values and interpret.

3.5. OLS forecast combination

Regress realised values on the two component forecasts (ARIMA and ETS) using the test-period observations:

$$Y_{t+h} = w_0 + w_1 \tilde{Y}_{t+h}^{\text{ARIMA}} + w_2 \tilde{Y}_{t+h}^{\text{ETS}} + e_t.$$

Report $\hat{w}_1$ and $\hat{w}_2$. Does the OLS combination improve on equal weights? Discuss why in small test samples equal weights often outperform OLS-optimal weights.

3.6. Business interpretation

Suppose you are the demand planner for this brand. Based on your forecast evaluation:

- (a) Which model would you adopt for production planning (MAPE $\leq 5\%$ is the typical industry threshold)?
- (b) How would you incorporate a known promotional event (e.g., Black Friday) into your ARIMA model?
- (c) What are the limitations of purely statistical forecasting for a brand subject to marketing interventions?

4. Core References

Diebold, F.X. & Mariano, R.S. (1995). *JBES*, 13, 253–263. Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2018). *PLOS ONE*, 13, e0194889. Hyndman, R.J. & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*, 3rd ed. <https://otexts.com/fpp3/>

Tutorial 7 — FGV EAESP Time Series Econometrics for Business Research.